

2. The concentration of glucose in the blood is normally between 60 mg and 160 mg per 100 cm³ of blood. Insulin reduces the concentration and prevents it from rising above the normal level.

(a) State the name of the organ which produces insulin. [1]

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(b) Doctors investigated the concentration of blood glucose in Kate and John. They suspected that Kate had diabetes but they knew that John did not.

Their blood glucose was measured after taking a glucose drink.
The results for John are shown in the table below. Kate's results are shown on the graph.

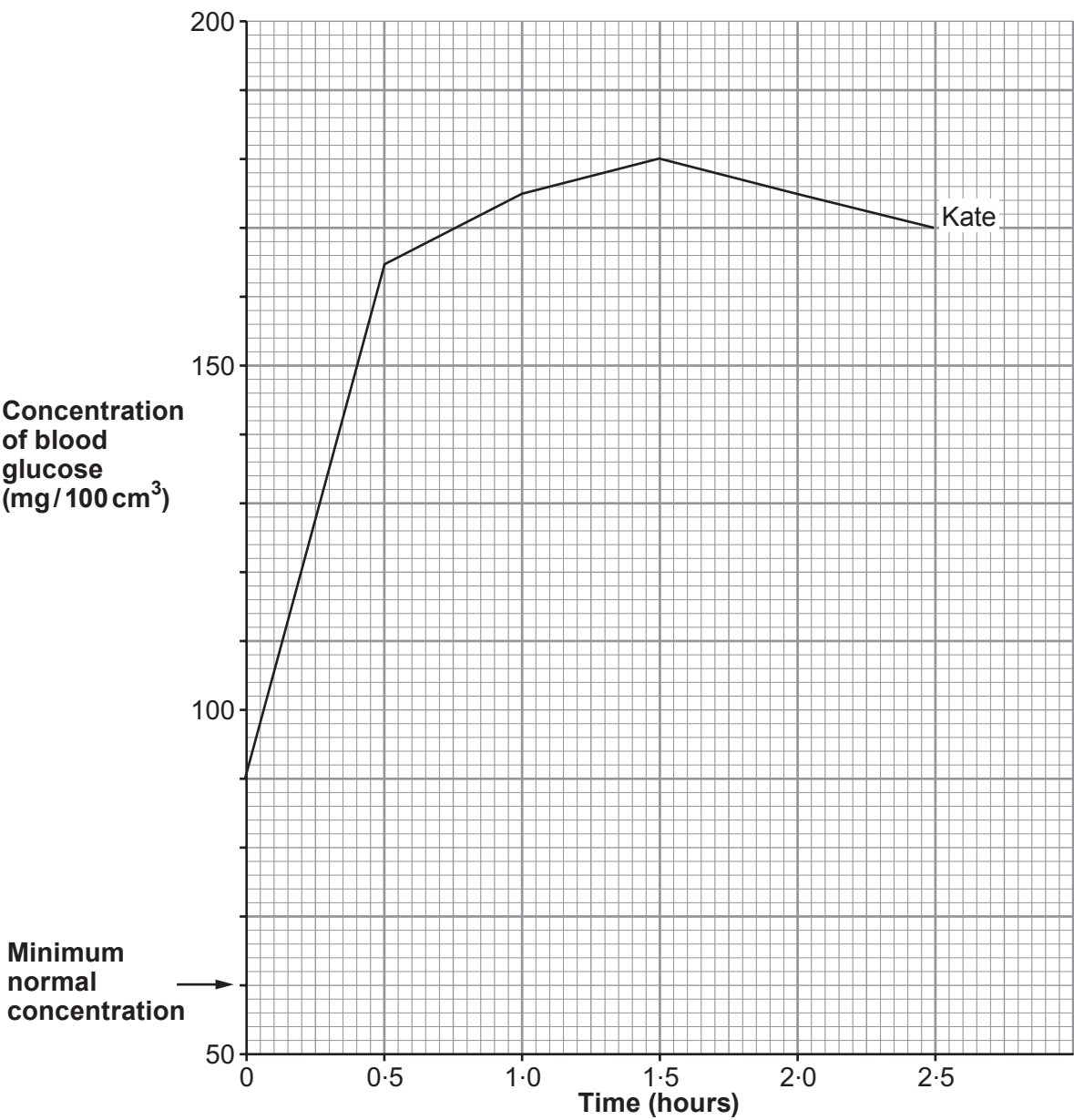
Time (hours)	John's blood glucose level (mg / 100 cm ³)
0	80
0.5	115
1.0	134
1.5	110
2.0	95
2.5	84

(i) Complete the graph of results by: [4]

- I. drawing an arrow on the axis for glucose concentration to show the **maximum normal concentration** of blood glucose (the minimum has been done for you);
- II. plotting the blood glucose results for John;
- III. joining your plots with a ruler and labelling your line.



Graph of results



Use the graph to answer the questions.

- (ii) At what time does insulin start to affect the concentration of glucose in John's blood? Give a reason for your answer. [2]

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Examiner
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(iii) How do the results for Kate at 0.5 hours show that she has diabetes? [1]

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(iv) Describe how the results for Kate are different from those of John, between 0.5 and 2 hours. [3]

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(v) The doctors concluded that Kate had diabetes. How could they increase the confidence they had in their results? [1]

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(c) (i) State **one** way in which Kate’s diabetes could be treated. [1]

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(ii) Arthur is 70 years old. He produces insulin but his liver cells do not respond to it. State the precise name of this medical condition. [1]

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